EXPLORATORY DATA ANALYSIS

CAPESTONE PROJECT

bigbasket

INTRODUCTION

In today's data-driven world, understanding customer preferences, product performance, and pricing strategies is essential for businesses to remain competitive. This project focuses on performing **Exploratory Data Analysis (EDA)** on a retail product dataset that includes information such as product name, category, sub-category, brand, pricing details, ratings, and descriptions.

The dataset consists of various products across multiple categories like **Beauty & Hygiene, Kitchen, Garden & Pets, and Cleaning & Household**, providing a comprehensive view of different consumer goods. Key attributes such as **sale price and market price** help in analysing pricing strategies and discounts, while **ratings** offer insights into customer satisfaction and product quality.

The main objective of this analysis is to:

- Understand the distribution of products across different categories and brands
- Analyse pricing patterns and identify price variations or discounts
- Examine customer ratings to evaluate product performance
- Identify trends, patterns, and potential outliers in the dataset

By applying various data visualization techniques such as **bar charts, histograms, and box plots**, this project aims to uncover meaningful insights that can help businesses improve decision-making, optimize pricing, and enhance customer satisfaction.

Overall, this EDA project serves as a foundation for deeper analysis and potential predictive modelling in the future.

OBJECTIVES

To analyze product distribution across categories and sub-categories

This objective focuses on understanding how products are divided among different categories such as Beauty & Hygiene, Kitchen, and Cleaning & Household. By analysing this distribution, we can identify which category has the highest number of products and which ones are less represented. This helps in understanding product variety and business focus areas.

To evaluate pricing patterns and discounts

In this objective, we compare the sale price with the market price to identify discounts and pricing strategies. This analysis helps in understanding how much price variation exists across products and whether certain categories or brands offer more discounts. It also helps detect overpriced or underpriced products.

To analyse customer ratings and product performance

This objective aims to study product ratings to evaluate customer satisfaction. By analysing rating distributions, we can identify high-performing and low-performing products. It also helps in understanding whether higher-priced products receive better ratings or not.

To identify top-performing brands and products

This involves analysing which brands have better product ratings, competitive pricing, and higher presence in the dataset. It helps in recognizing popular and trusted brands as well as those that may need improvement in quality or pricing.

To detect trends, patterns, and outliers in the dataset

The goal here is to use visualizations like histograms, box plots, and bar charts to uncover hidden patterns in the data. This includes identifying unusual values (outliers), trends in pricing or ratings, and any inconsistencies that may affect analysis or decision-making.

IMPORTANT PYTHON LIBRARIES

Pandas (import pandas as pd)

Pandas is used for data manipulation and analysis. It provides data structures like Data Frame which make it easy to handle structured data. In this project, Pandas is used for:

- Loading the dataset from CSV/Excel files
- Cleaning the data (handling missing values, duplicates)
- Filtering and grouping data based on category, brand, etc.
- Comparing sale price and market price

Pandas is the core library used throughout the project for data analysis.

NumPy (import numpy as np)

NumPy (Numerical Python) is used for performing mathematical and numerical operations efficiently. It provides support for arrays and statistical calculations. In this project, NumPy is used for:

- Performing calculations like mean, median, and standard deviation
- Supporting data manipulation operations in Pandas
- Handling numerical computations efficiently

Although most operations are done using Pandas, NumPy works in the background to make data processing faster.

Matplotlib (import matplotlib.pyplot as plt)

Matplotlib is a basic data visualization library used to create charts and graphs. It helps in understanding data through visual representation. In this project, Matplotlib is used for:

- Creating histograms to show distribution of prices and ratings
- Plotting bar charts for category-wise product count
- Visualizing trends and patterns in the dataset

It provides a strong foundation for building visual insights.

Seaborn (import seaborn as sns)

Seaborn is an advanced visualization library built on top of Matplotlib. It provides more attractive and informative statistical graphics. In this project, Seaborn is used for:

- Creating box plots to detect outliers in prices
- Generating count plots for category and brand distribution
- Creating heatmaps to understand relationships between variables

Seaborn makes the graphs more visually appealing and easier to interpret.

Warnings (import warnings)

The warnings library is used to ignore unnecessary warning messages during execution. In this project, it is used for:

- Suppressing unwanted warnings
- Keeping the output clean and readable

This improves the overall presentation of the results.

DATA EXPLORATION

Loading Dataset

```
#Load Dataset
import pandas as pd
df = pd.read_csv("C:/Users/yadav/Downloads/BigBasket Products.csv")
```

I have loaded the **BigBasket product dataset** into the Python environment using the Pandas library. The pd.read_csv() function is used to import the dataset from the specified file path and store it in a DataFrame named **df**, which serves as the primary data structure for further analysis. The dataset contains important columns such as **product, category, sub_category, brand, sale_price, market_price, type, rating, and description**, which provide detailed information about different products.

Pandas enables efficient handling of structured data and provides flexible tools for data cleaning, manipulation, and exploration. This allows easy analysis of product distribution, pricing patterns, and customer ratings. This step forms the foundation of the project, ensuring that the dataset is successfully loaded and ready for exploratory data analysis (EDA).

Basic Composition of Data

df

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.00	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.00	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.00	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.00	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi...

The dataset contains information about **BigBasket products**. It has around **27,555 rows and 10 columns**, where each row represents a unique product item. It includes details such as **product name, category, sub-category, brand, sale price, market price, type, rating, and description**.

Most of the columns are **categorical** in nature, such as product, category, sub-category, brand, type, and description, while **sale_price, market_price, and rating** are numerical columns.

Some columns like **brand, rating, and description** may contain missing values, which need to be handled during the data cleaning process to ensure accurate analysis.

DATA EXPLORATION

Exploring the Dataset

```
df.head(12)
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
0	1	Garlic Oil - Vegetarian Capsule 500 mg	Beauty & Hygiene	Hair Care	Sri Sri Ayurveda	220.0	220.0	Hair Oil & Serum	4.1	This Product contains Garlic Oil that is known...
1	2	Water Bottle - Orange	Kitchen, Garden & Pets	Storage & Accessories	Mastercook	180.0	180.0	Water & Fridge Bottles	2.3	Each product is microwave safe (without lid), ...
2	3	Brass Angle Deep - Plain, No.2	Cleaning & Household	Pooja Needs	Trm	119.0	250.0	Lamp & Lamp Oil	3.4	A perfect gift for all occasions, be it your m...
3	4	Cereal Flip Lid Container/Storage Jar - Assort...	Cleaning & Household	Bins & Bathroom Ware	Nakoda	149.0	176.0	Laundry, Storage Baskets	3.7	Multipurpose container with an attractive desi...
4	5	Creme Soft Soap - For Hands & Body	Beauty & Hygiene	Bath & Hand Wash	Nivea	162.0	162.0	Bathing Bars & Soaps	4.4	Nivea Creme Soft Soap gives your skin the best...
5	6	Germ - Removal Multipurpose Wipes	Cleaning & Household	All Purpose Cleaners	Nature Protect	169.0	199.0	Disinfectant Spray & Cleaners	3.3	Stay protected from contamination with Multipu...
6	7	Multani Mati	Beauty & Hygiene	Skin Care	Satinance	58.0	58.0	Face Care	3.6	Satinance multani matti is an excellent skin t...
7	8	Hand Sanitizer - 70% Alcohol Base	Beauty & Hygiene	Bath & Hand Wash	Bionova	250.0	250.0	Hand Wash & Sanitizers	4.0	70%Alcohol based is gentle of hand leaves skin...
8	9	Biotin & Collagen Volumizing Hair Shampoo + Bi...	Beauty & Hygiene	Hair Care	StBotanica	1098.0	1098.0	Shampoo & Conditioner	3.5	An exclusive blend with Vitamin B7 Biotin, Hyd...
9	10	Scrub Pad - Anti- Bacterial, Regular	Cleaning & Household	Mops, Brushes & Scrubs	Scotch brite	20.0	20.0	Utensil Scrub-Pad, Glove	4.3	Scotch Brite Anti- Bacterial Scrub Pad thoroug...
10	11	Wheat Grass Powder - Raw	Gourmet & World Food	Cooking & Baking Needs	NUTRASHIL	261.0	290.0	Flours & Pre-Mixes	4.0	Wheatgrass is a superfood potent health food w...
11	12	Butter Cookies Gold Collection	Gourmet & World Food	Chocolates & Biscuits	Sapphire	600.0	600.0	Luxury Chocolates, Gifts	2.2	Enjoy a tin full of delicious butter cookies m...

I used the **df.head(12)** function to display the first 12 rows of the BigBasket dataset. This helped me to understand the structure and content of the dataset.

From the output, I found that the dataset contains columns such as **product, category, sub_category, brand, sale_price, market_price, type, rating, and description**. It also showed sample values for each column, which helped in verifying that the data has been loaded correctly.

I observed that different products belong to different categories like *Beauty & Hygiene*, *Kitchen, Garden & Pets*, and *Cleaning & Household*. The prices and ratings vary across products, indicating diversity in product range and customer feedback.

This step was useful for getting an initial overview of the dataset before performing further analysis.

DATA EXPLORATION

Statistical Description of the Dataset

```
df.describe()
```

	index	sale_price	market_price	rating
count	27555.00000	27549.000000	27555.000000	18919.000000
mean	13778.00000	334.648391	382.056664	3.943295
std	7954.58767	1202.102113	581.730717	0.739217
min	1.00000	2.450000	3.000000	1.000000
25%	6889.50000	95.000000	100.000000	3.700000
50%	13778.00000	190.320000	220.000000	4.100000
75%	20666.50000	359.000000	425.000000	4.300000
max	27555.00000	112475.000000	12500.000000	5.000000

I used the **df.describe()** function to generate the statistical summary of the numerical columns in the Big Basket dataset. This function provides key metrics such as **count, mean, standard deviation, minimum value, maximum value, and quartiles (25%, 50%, 75%)**.

From the output, I found that the dataset contains numerical columns like **sale_price, market_price, and rating**. The summary helped me understand the average price of products, the variation in pricing, and the distribution of customer ratings.

I also observed the range of values, which helped in identifying whether there are any extreme values or inconsistencies in the data. This step is useful for gaining insights into the overall distribution and behaviour of numerical variables before performing further analysis.

DATA EXPLORATION

Understanding Data Structure

```
df.info()

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 27555 entries, 0 to 27554
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype
---  ------        --------------  -----
 0   index         27555 non-null  int64
 1   product       27554 non-null  object
 2   category      27555 non-null  object
 3   sub_category  27555 non-null  object
 4   brand         27554 non-null  object
 5   sale_price    27549 non-null  float64
 6   market_price  27555 non-null  float64
 7   type          27555 non-null  object
 8   rating        18919 non-null  float64
 9   description   27440 non-null  object
dtypes: float64(3), int64(1), object(6)
memory usage: 2.1+ MB
```

I used the **df.info()** function to get detailed information about the structure of the Big Basket dataset. This function provides a summary that includes the total number of rows and columns, column names, data types, and the number of non-null values in each column.

From the output, I found that the dataset contains **27,555 entries and 10 columns**. It also showed that columns like **product, category, sub category, brand, type, and description** are of object (categorical) data type, while **sale price, market price, and rating** are numerical.

I also observed that some columns may contain missing values, as indicated by fewer non-null counts. This step helped me identify which columns require data cleaning and gave a clear understanding of the dataset structure before moving to further analysis.

DATA ANALYSIS

Identifying Top Performing Products

```
top_products = df.sort_values(by='rating', ascending=False).head(5)
top_products
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
8987	8988	Bernard Scot Ice Bucket With Tong & Glasses/Tu...	Kitchen, Garden & Pets	Crockery & Cutlery	Treo	639.0	790.0	Glassware	5.0	Now spend some relaxing time at home or with y...
8267	8268	Skin-Fit Tablets - Improves Skin Health	Beauty & Hygiene	Health & Medicine	Jiva Ayurveda	63.0	63.0	Everyday Medicine	5.0	The Jiva Skin Fit tablet aims at keeping our s...
19459	19460	Disposable Face Mask - Melt Blown, 3 Ply	Beauty & Hygiene	Health & Medicine	Mirabelle	49.0	119.0	Face Masks & Safety Gears	5.0	We can use the mask multiple times till it get...
19447	19448	Caviar Rejuvenate & Protecting Face Mask Sheet	Beauty & Hygiene	Skin Care	Glamveda	120.0	150.0	Face Care	5.0	Moist & elastic skin with nutritious caviar. G...
25246	25247	Adult Dry Dog Food - Chicken & Rice	Kitchen, Garden & Pets	Pet Food & Accessories	Chappi	1060.2	1140.0	Pet Meals & Treats	5.0	Chappi is complete and balanced nutrition for ...

I sorted the dataset in descending order based on the **rating** column to identify the top-performing products. Since the dataset does not include sales quantity, **ratings are used as an indicator of product performance**.

From the output, I found that the top products have the highest ratings, which indicates that these products are highly preferred by customers. These products reflect better quality, higher satisfaction, and strong customer trust.

This analysis helps in identifying the best products that can be promoted further to increase sales and customer engagement.

Identifying Least Performing Products

```
least_products = df.sort_values(by='rating', ascending=True).head(5)
least_products
```

	index	product	category	sub_category	brand	sale_price	market_price	type	rating	description
24845	24846	Divine-Issme Set - Eau De Parfum + Deodorant	Beauty & Hygiene	Fragrances & Deos	Ulric De Varens	918.75	1225.0	Gift Sets	1.0	Naturally masculine, lavender enhances this ca...
5731	5732	Cooling Teether - Trumpet	Baby Care	Baby Accessories	Pigeon Baby	279.00	279.0	Soothers & Teethers	1.0	This cooling teether is designed to fit into y...
5677	5678	Retinol Advanced Anti Aging Day Cream - SPF 30...	Beauty & Hygiene	Skin Care	StBotanica	899.00	899.0	Face Care	1.0	St.Botanica Retinol Advanced anti-ageing Day C...
17284	17285	Soap Case - Avenue Nxt	Beauty & Hygiene	Bath & Hand Wash	Plasto World	95.00	95.0	Bathing Accessories	1.0	OurÂ Soap DishÂ is made of plastic infused wit...
10453	10454	Milk Chocolate - Salted Peanut Praline	Snacks & Branded Foods	Chocolates & Candies	Uns	295.00	295.0	Chocolates	1.0	With a taste that lingers on, enjoy smooth 36%...

I sorted the dataset in ascending order based on the **rating** column to identify the least-performing products.

From the output, I found that the least performing products have the lowest ratings, which may indicate poor customer satisfaction, quality issues, or lack of popularity among users.

This step is important as it helps in identifying products that may require improvement in quality, pricing, or marketing strategies to enhance their performance.

DATA ANALYSIS

Measuring Discount on Products

```
df['discount'] = df['market_price'] - df['sale_price']

df[['product', 'market_price', 'sale_price', 'discount']].head()
```

	product	market_price	sale_price	discount
0	Garlic Oil - Vegetarian Capsule 500 mg	220.0	220.0	0.0
1	Water Bottle - Orange	180.0	180.0	0.0
2	Brass Angle Deep - Plain, No.2	250.0	119.0	131.0
3	Cereal Flip Lid Container/Storage Jar - Assort...	176.0	149.0	27.0
4	Creme Soft Soap - For Hands & Body	162.0	162.0	0.0

I calculated the discount on each product by subtracting the **sale_price** from the **market_price** and created a new column named **discount** in the dataset.

From the output, I found that some products have no discount (where sale price is equal to market price), while others have a significant difference between market price and sale price, indicating the discount offered.

This analysis helps in identifying which products are offered at higher discounts and can attract more customers. It also provides insights into pricing strategies followed by different brands and categories.

Identifying Missing Values in the Dataset

```
df.isnull().sum()

index            0
product          1
category         0
sub_category     0
brand            1
sale_price       6
market_price     0
type             0
rating        8636
description     115
discount          6
dtype: int64
```

I used the **isnull().sum()** function to identify missing values in each column of the Big Basket dataset. This function returns the total number of null (missing) values present in every column.

From the output, I found that some columns like **brand, rating, or description** may contain missing values, while other columns may have complete data. Identifying missing values is an important step before performing analysis, as missing data can affect the accuracy of results.

This step helps in deciding how to handle missing values, such as removing them or filling them with appropriate values (mean, median, or mode), to ensure better data quality and reliable analysis.

DATA ANALYSIS

Detecting and Handling Outliers in the Dataset

```
import numpy as np
def handle_outliers(column):
    Q1 = df[column].quantile(0.25)
    Q3 = df[column].quantile(0.75)
    IQR = Q3 - Q1

    lower_bound = Q1 - 1.5 * IQR
    upper_bound = Q3 + 1.5 * IQR


    mean_value = df[column].mean()
    df[column] = np.where((df[column] < lower_bound) | (df[column] > upper_bound),
                          mean_value,
                          df[column])
handle_outliers('sale_price')
handle_outliers('market_price')
handle_outliers('rating')
```

I used the **IQR (Interquartile Range) method** to detect outliers in numerical columns such as **sale_price, market_price, and rating**.

First, I calculated the first quartile (Q1) and third quartile (Q3), and then computed the IQR. Using this, I determined the lower and upper bounds to identify outliers. Any values falling outside these bounds were considered outliers.

After detecting the outliers, I replaced them with the **mean value** of the respective column. This helps in reducing the impact of extreme values and makes the dataset more balanced for analysis.

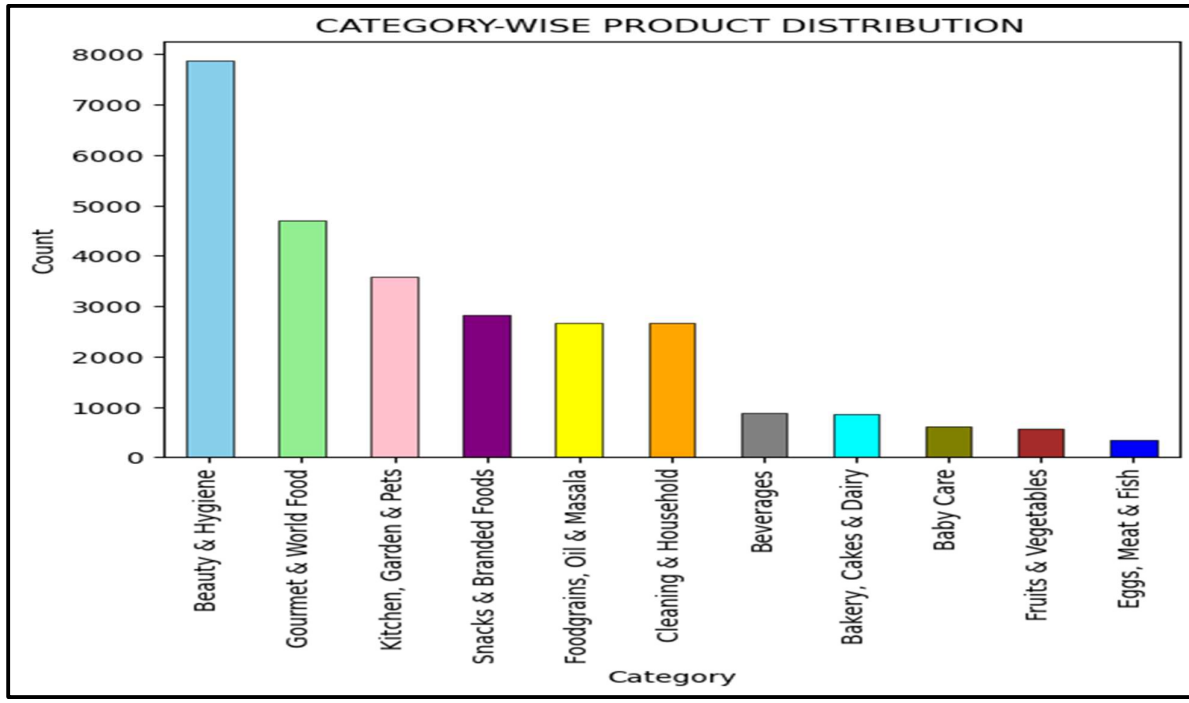
This step is important because outliers can distort the results and affect the accuracy of data analysis and visualization.

DATA VISUALISATION

Data Visualise Using Bar Chart

```
import matplotlib.pyplot as plt
import seaborn as sns

# 1. Category-wise product count
plt.figure()
colors = ['skyblue','lightgreen','pink','purple','yellow','orange','grey','cyan','olive','brown','blue']
df['category'].value_counts().plot(kind='bar',color = colors, edgecolor = 'black', linewidth = 0.5)
plt.title("CATEGORY-WISE PRODUCT DISTRIBUTION")
plt.xlabel("Category")
plt.ylabel("Count")
plt.show()
```

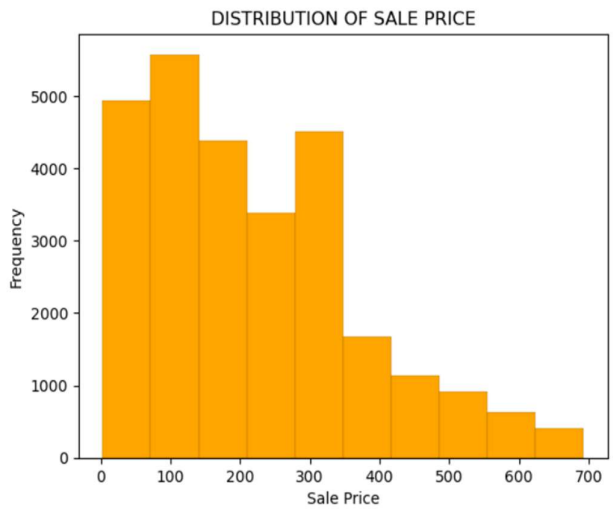


Firstly, I used a **bar chart** to represent the number of products in each category. This helped me understand which category has the highest and lowest number of products. It gives a clear picture of product distribution across categories like Beauty & Hygiene, Kitchen, and Cleaning.

DATA VISUALISATION

Data Visualise Using Histogram

```
# 2. Distribution of Sale Price
plt.figure()
color = 'orange'
plt.hist(df['sale_price'], color = color, edgecolor = 'black', linewidth = 0.1)
plt.title("DISTRIBUTION OF SALE PRICE")
plt.xlabel("Sale Price")
plt.ylabel("Frequency")
plt.show()
```

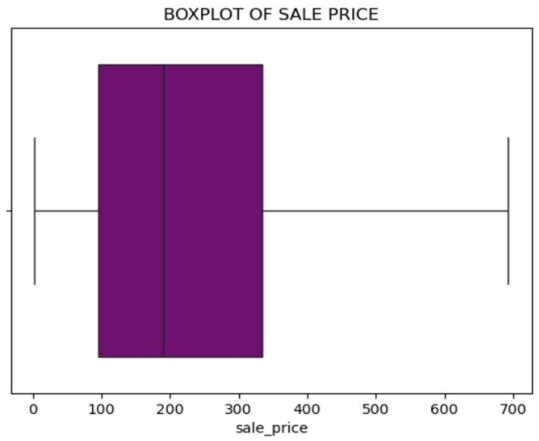


Secondly, I created a **histogram** for the **sale_price** column. This helped me analyse how product prices are distributed. From this, I could observe whether most products fall in a lower price range or higher price range, and whether the data is skewed.

DATA VISUALISATION

Data Visualise Using Boxplot

```
# 3. Boxplot for detecting outliers in price
plt.figure()
color = 'purple'
sns.boxplot(x=df['sale_price'], color = color)
plt.title("BOXPLOT OF SALE PRICE")
plt.show()
```

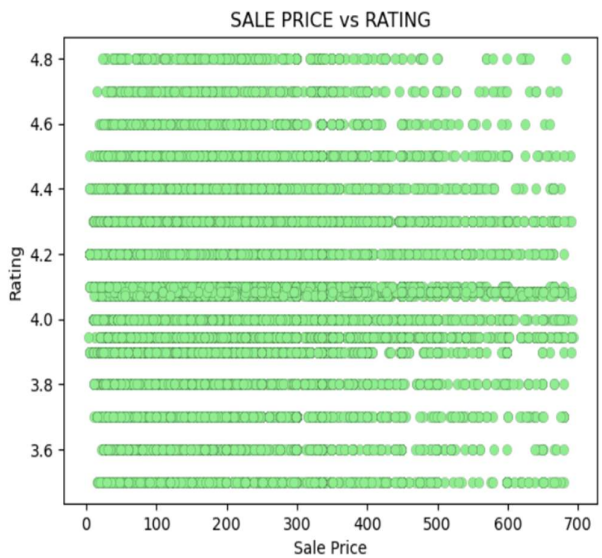


Next, I used a **boxplot** to detect outliers in the sale price. The boxplot visually shows the spread of the data and highlights any extreme values that lie far from the normal range. This helps in identifying unusual pricing in the dataset.

DATA VISUALISATION

Data Visualise Using Scatter Plot

```python
# 4. Scatter plot for Price vs Rating
plt.figure()
color = ['lightgreen']
plt.scatter(df['sale_price'], df['rating'], color = color, edgecolor = 'black', linewidth = 0.1)
plt.title("SALE PRICE vs RATING")
plt.xlabel("Sale Price")
plt.ylabel("Rating")
plt.show()
```



Finally, I created a **scatter plot** between **sale_price and rating**. This helped me understand the relationship between product price and customer ratings. From this, I could observe whether higher-priced products tend to have better ratings or if there is no strong relationship.

Overall, these visualizations made it easier to interpret the dataset and provided meaningful insights that are not easily visible in raw data.

RECOMMENDATION

Focus on High-Rated Products

Products with higher ratings indicate better customer satisfaction and quality. These products should be promoted more through advertisements and featured listings. Highlighting top-rated products can increase customer trust and boost overall sales.

Improve Low-Rated Products

Products with low ratings may have issues related to quality, pricing, or customer expectations. Businesses should analyze customer feedback and take corrective actions such as improving product quality, updating descriptions, or revising pricing strategies.

Optimize Pricing Strategy

From the analysis of sale price and market price, it is recommended to maintain competitive pricing. Products with reasonable discounts tend to attract more customers. Businesses should ensure that pricing is aligned with customer expectations and market trends.

Leverage Discounts and Offers

Products with higher discounts can attract more buyers. It is recommended to strategically provide discounts on selected products to increase sales volume, especially in competitive categories.

Focus on Popular Categories

Categories with a higher number of products and better ratings should be given more focus. Expanding product variety in these categories can help in increasing customer engagement and overall business growth.

Focus on Popular Categories

Clear and detailed product descriptions help customers make better purchasing decisions. It is recommended to improve product descriptions and ensure proper branding for all items. Well-described products with complete information can increase customer confidence and reduce confusion, ultimately improving sales and ratings.

CONCLUSION

Diverse Product Range

The dataset shows that Big Basket offers a wide variety of products across multiple categories, which helps in catering to different customer needs and preferences.

Variation in Pricing

There is a significant variation between sale price and market price, indicating that discounts and pricing strategies play an important role in attracting customers.

Customer Ratings Provide Key Insights

Ratings help in understanding product performance. High-rated products reflect customer satisfaction, while low-rated products highlight areas needing improvement.

Presence of Data Issues

The dataset contains missing values and outliers, which required cleaning. Proper data preprocessing is essential to ensure accurate and meaningful analysis.

Visualizations Enhance Understanding

Using charts and graphs made it easier to identify patterns, trends, and relationships in the dataset, such as price distribution and the relationship between price and ratings.

Brand Influence on Product Performance

The analysis indicates that certain brands perform better in terms of ratings and pricing. This shows that brand reputation plays an important role in influencing customer choices and product success.

Relationship Between Price and Rating

From the scatter plot analysis, it can be observed that there is no strong direct relationship between price and rating. This means that higher-priced products do not always guarantee better customer satisfaction, highlighting the importance of quality over price.

THANKS FOR READING

For Code:

https://github.com/shwetayadav376-spec/Big-Basket-EDA-Project